Open Local AI Research Node

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Private, reproducible local compute for AI-for-science and AI-safety research — with ASRA, a neuro-symbolic system that learns world dynamics by intervening.

THE GAP

Open models do not create an independent research ecosystem. Serious work still depends on centralized APIs and rented GPUs: recurring cost, privacy exposure, rate limits, and experiments that are hard to reproduce. Buying a GPU does not close the gap. Every lab still has to assemble drivers, containers, model serving, storage, authentication, scheduling, monitoring, backups, and recovery.

WHAT I AM BUILDING

The Node. A secure, affordable local platform for private open-model inference, bounded fine-tuning, long-running multi-agent workflows, and reproducible experiments. Open-source install and management software, at least two validated hardware configurations, a secure-by-default reference architecture, and automated health checks, monitoring, backup, and recovery.

ASRA. The flagship application. It learns world dynamics through intervention, keeps discovered mechanisms explicit, and plans under uncertainty. Tabby inquires, Grubby handles evidence, and Sam decides. They observe, hypothesize, intervene, evaluate, and update an explicit world model — contradictions, predictions, uncertainty, and an evidence trail.

WHY LOCAL COMPUTE

High-frequency experiments without per-call API cost. Unpublished traces stay on hardware the researcher owns. Persistent scientific memory, lower latency, and direct model inspection. Independent safety evaluation that does not depend entirely on a frontier lab's infrastructure, policies, or uptime.

12 MONTHS

Starts within one week of funding. Compare ASRA with single-agent and unstructured multi-agent baselines on task success, hypothesis quality, calibration, sample efficiency, contradiction recovery, and transfer — plus latency, energy, cost, reliability, and reproducibility. First settings: ARC-AGI-3 and synthetic interactive environments, then one bounded scientific use case on a licensed public dataset. Pilot with about 5–10 independent researchers.

Months 1–2	Requirements, security model, hardware configs, experiment design, baselines
Months 3–4	First node: automated install, private model serving, storage, auth, monitoring
Months 5–7	ASRA multi-agent system: hypotheses, interventions, evidence, uncertainty, world-model updates
Months 8–9	Benchmark two hardware configurations against baselines; reproducibility, energy, and cost
Months 10–11	Pilot, security review, one public-data scientific use case
Month 12	Open-source release, hardware guides, results, pilot findings, federation roadmap

Three-year path. One validated node becomes a trusted network of independently operated nodes at universities, nonprofits, community labs, and independent researchers. Coding agents can take on deployment, checks, and diagnostics. Work stays bounded by isolation, permissions, audit logs, and human oversight — independent evaluation and shared safety benchmarks outside any single lab or cloud.

FOUNDER

About 20 years building production systems across machine learning, distributed systems, cloud infrastructure, and enterprise platforms, including Fortune 500 work. Independently built AI Tutor, a deployed multi-agent learning platform, and ASRA/NFM, an ongoing neuro-symbolic research program. Published U.S. patent application US 2024/0273653 A1.

THE ASK

$95,000 for 12 months. Lead engineering and research; GPU workstations, storage, networking, and measurement equipment; targeted security review; pilot support and Bay Area working sessions; documentation. Project-funded software, hardware guides, evaluation methods, and non-sensitive artifacts will be open-sourced. Services stay private by default, with authentication, isolation, and audit logs. Seeking accelerator partners and early investors for this build.

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